



A Handy Optical Wattmeter

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All residential and industrial units have energy meters that show energy consumed by the users. But the power (watts) is not shown by these meters, except by some new models.

During measurement, all energy meters flash an LED at a predetermined rate that is called impulse rate. The common flash rates are 6,400 impulses/kWh, 3,200 impulses/

kWh, etc. By using the flash rate and some math, the power can also be calculated.

Presented here is a circuit to calculate the power consumed with the help of pulsing LED light of the energy meter and show it.

Circuit and working

Circuit diagram of the optical wattmeter is shown in Fig. 1. The cir-

PARTS LIST

Semiconductors:

- Board1 - Arduino Nano
- LCD1 - 16x2 LCD
- T1 - BC547 NPN transistor
- T2 - L14F1 photo-transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 270-ohm
- R2 - 10-kilo-ohm
- R3 - 1-kilo-ohm
- VR1 - 10-kilo-ohm potmeter

Miscellaneous:

- Jumper wires
- 5V power supply

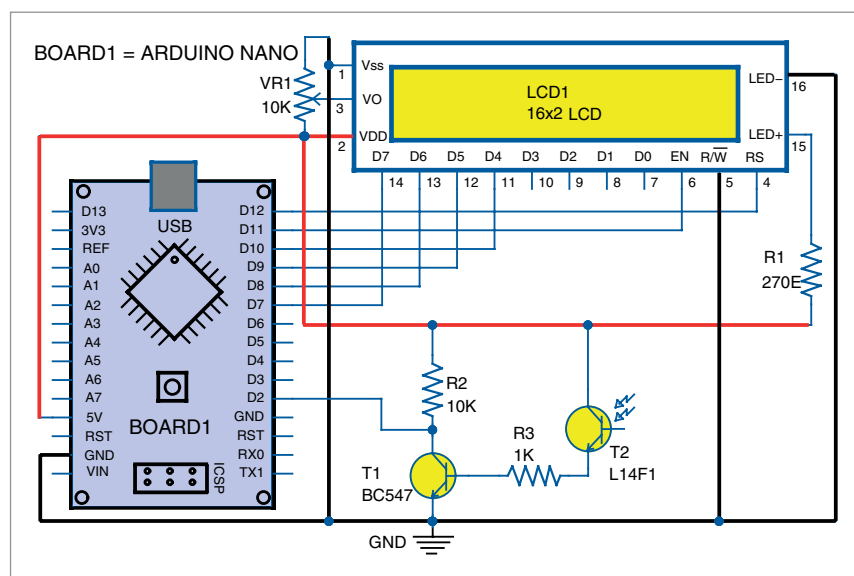


Fig. 1: Circuit diagram

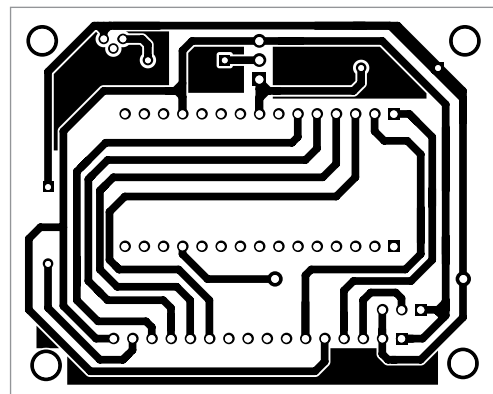


Fig. 2: Actual-size PCB layout

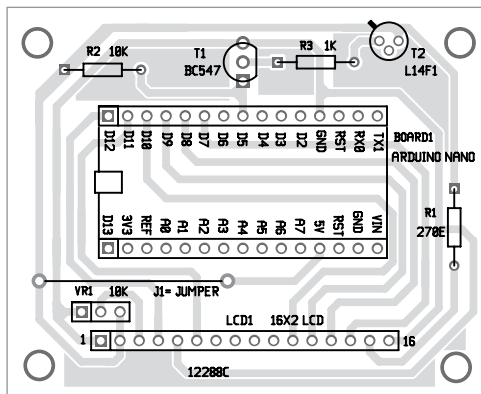


Fig. 3: Component layout of the PCB

cuit includes Arduino Nano board (Board1), 16x2 LCD display LCD1, NPN transistor BC547 (T1), and photo-transistor L14F1 (T2).

Whenever the energy meter's LED light falls on photo-transistor T2, it conducts, thereby activating transistor T1. The pulse available at collector of T1 is given to first interrupt pin (pin D2) of the Arduino board. The Arduino calculates the power consumed in watts with the help of software, which is updated to the LCD display simultaneously.

Software

Arduino IDE is used to compile and upload program *OpticalWattmeter.ino* to the Arduino Nano board. Select

the correct board from Board→Tools menu in Arduino IDE, select COM port and upload the program through the standard USB port in computer.

Note. There may be deviation in the wattage readings due to the use of millis library. If more accurate result is needed, use an ATmega328 chip with external crystal instead of resonator in the Arduino board and use the inbuilt hardware timer.